Homework:

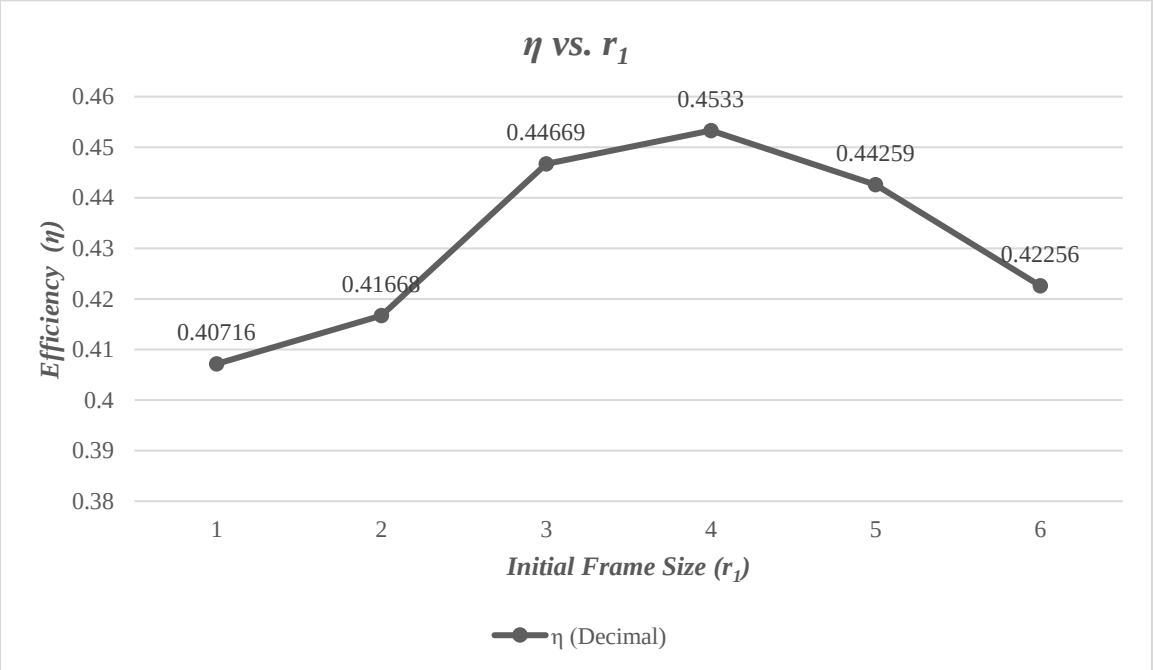
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Exercise Question 3:

1. Find the overall collision resolution efficiency η in the different cases in which the initial frame size is set to r1=1,2,3,4,5,6

ANSWER:

r_1	L_4^* (Fraction)	L_4^* (Decimal)	η (Fraction)	η (Decimal)
1	1061/108	9.82407	432/1061	0.40716
2	4147/432	9.59953	1728/4147	0.41668
3	26111/2916	8.95439	11664/26111	0.44671
4	953/108	8.82407	432/953	0.45330
5	122009/13500	9.03770	54000/122009	0.44259
6	110413/11664	9.46605	46656/110413	0.42256



Comment on Results:

- Highest efficiency **η** is achieved when ***Number of Tags (N) = Initial Frame Size (r_1)***.
  - Smaller r_1 means higher probability of collision, but less slots are left empty.
  - Larger r_1 means lower probability of collision, but some of the slots are left empty and are wasted.

Answer motivation:

- Given the recursive formula when $n = r_1$

$$L_n = \frac{n + \sum_1^{n-1} P(s = i) \times L_{n-i}}{\left(1 \; - \; P(s = 0)\right)}$$

- And the general formula when $n \neq r1$

$$L_n^* = \; r_1 \; + \sum_{i \, = \, 0}^{n} P(s = i \mid r_1) \times L_{n-i}$$

- In this report I frequently used the formula $C(n, k)$ to quickly find all possible permutations

$$C(n, k) = \frac{n!}{k! \times (n - k)!}$$

Step 1: Calculating L_n (Optimal Resolution Times) for n = 0, 1, 2, 3, 4
These values are needed for the $\boldsymbol{L_{n-i}}$ terms in Formula 2.

- $\boldsymbol{L_0 = \; 0}$
- $\boldsymbol{L_1 = \; 1}$
- $\boldsymbol{L_2 = \; 4}$ (Given)
- $\boldsymbol{L_3 = \frac{51}{8} = \; 6.375}$ (Given)
- $\boldsymbol{L_4 = \frac{953}{108} \approx 8.8241}$ (Calculations detailed below)

Step 2: Calculate L_N^* for n = 4 and each r_1

A. For $r_1 = 1$ (N=4 tags, L = 1 slot)

- **Probability Calculations $P(S = i \mid N = 4, L = 1)$:**
  - The only possible outcome is that all 4 tags choose the single slot.
    - Pattern: (4) in the slot.
    - Successes s = 0 (collision).
  - Therefore:
    - $P(S = 0 \mid 4,1) = 1/1 = 1$
    - $P(S = 1 \mid 4,1) = 0$
    - $P(S = 2 \mid 4,1) = 0$
    - $P(S = 3 \mid 4,1) = 0$
    - $P(S = 4 \mid 4,1) = 0$

$$L_n^* = 1 + [P(s = 0 \mid 4,1) \cdot L_4] + \cdots$$

$$L_4^* = 1 + \left[1 \cdot \left(\frac{953}{108}\right)\right] = \frac{1061}{108} \approx 9.8241$$

B. For $r_1 = 2$ (N = 4 tags, L = 2 slots)

- **Probability Calculations $P(s = i \mid N = 4, L = 2)$:**
  - Total ways for 4 tags to choose 2 slots = $\mathbf{2^4}$ = 16.
  - Patterns leading to s = 0 successes:
    1. (4,0): All 4 tags in one slot, 1 slot empty.
       - Choose a slot for all 4 tags:
         - C(2,1) = 2 possible unique slot selections for the only pair.
    2. (2,2): 2 tags in one slot, 2 tags in the other.
       - Choose 2 tags for the first slot:
         - C(4,2) = 6 possible unique pairs (The remaining 2 go to the other slot)
    - Total ways for $(s = 0) = 2 + 6 = 8$.
  - **Patterns leading to s=1 success:**

1. (3,1): 3 tags in one slot (collision), 1 tag in the other (success).
    - Choose 3 tags for the collision group: $\boldsymbol{C(4,3) = 4}$.
    - Choose slot for the 3 tags: $\boldsymbol{C(2,1) = 2}$.
        - The singleton goes in the other slot.
    - $\boldsymbol{Ways = 4 \times 2 = 8}$

- **Patterns leading to $\boldsymbol{s = 2}$ successes:**
    - With 4 tags and 2 slots it is impossible to have two distinct successes. *Ways = 0.*
- Probabilities:
    - $\boldsymbol{P(S = 0 \mid 4,2) = 8/16 = 1/2}$
    - $\boldsymbol{P(S = 1 \mid 4,2) = 8/16 = 1/2}$
    - $\boldsymbol{P(S = 2 \mid 4,2) = 0/16 = 0}$
    - $\boldsymbol{P(S = 3 \mid 4,2) = 0}$
    - $\boldsymbol{P(S = 4 \mid 4,2) = 0}$

- **Calculate $\boldsymbol{L_4^*}$ $\boldsymbol{given}$ $\boldsymbol{(r_1 = 2)}$:**

$$\boldsymbol{L_4^* = 2 + [P(S = 0|4,2) \cdot L_4] + [P(S = 1|4,2) \cdot L_3] + \ldots}$$

$$\boldsymbol{L_4^* = 2 + \left[\left(\frac{1}{2}\right) \cdot \left(\frac{953}{108}\right)\right] + \left[\left(\frac{1}{2}\right) \cdot \left(\frac{51}{8}\right)\right] = \frac{4147}{432} \approx 9.5995}$$

C. For $r_1 = 3$ $(N = 4\ tags, L = 3\ slots)$

- **Probability Calculations $\boldsymbol{P}$ $\boldsymbol{(s = i \mid N = 4, L = 3)}$:**
    - Total ways = $\boldsymbol{3^4}$ = 81.
    - $\boldsymbol{s = 0}$ **successes:**
        1. (4,0,0): All 4 in one slot.
            - $\boldsymbol{C(3,1) = 3}$ ways to slot the pair
        2. (2,2,0): Two pairs in two slots.
            - $C(4,2) = 6$ possible pairings
            - $C(3,2) = 3$ possible slot locations
            - $Total = 6 \cdot 3 = 18$ ways
        - Total for $\boldsymbol{(s = 0) = 3 + 18 = 21}$ ways

- **$s = 1$ success:**
    1. (3,1,0): 3 in one, 1 in another.
        - $C(4,3) = 4$ possible pairings
        - $C(3,1) = 3$ possible locations for the pairing
        - $C(2,1) = 2$ possible locations for the singleton
        - $4 \cdot 3 \cdot 2 = 24\ ways$.
- **$s = 2$ successes:**
    1. (2,1,1): Pair in one, two singletons.
        - $C(4,2) = 6$ possible pairings
        - $C(3,1) = 3$ possible locations for the pair
            - *Remaining 2 slots for 2 singletons*
        - $2! = 2$ possible ways to arrange the singletons in the slots
        - $6 \cdot 3 \cdot 2 = 36$ ways to get $s = 2$
- Probabilities:
    - $P(S = 0 \mid 4,3) = 21/81$
    - $P(S = 1 \mid 4,3) = 24/81$
    - $P(S = 2 \mid 4,3) = 36/81$
    - $P(S = 3 \mid 4,3) = 0$
    - $P(S = 4 \mid 4,3) = 0$

- **Calculate L_4^* for ($r_1 = 3$):**

$$L_4^* = 3 + \left[\left(\frac{21}{81}\right)L_4\right] + \left[\left(\frac{24}{81}\right)L_3\right] + \left[\left(\frac{36}{81}\right)L_2\right]$$

$$L_4^* = 3 + \left(\frac{21}{81}\right)\left(\frac{953}{108}\right) + \left(\frac{24}{81}\right)\left(\frac{51}{8}\right) + \left(\frac{36}{81}\right)(4) = \frac{26111}{2916} \approx 8.9544$$

D. For $r_1 = 4$ ($N = 4\ tags, L = 4\ slots$)

- **Probability Calculations $P(s = i \mid N = 4, L = 4)$:**
    - Total ways = $4^4 = 256$.

- **$s = 0$ successes:**
    1. (4,0,0,0):
        - $\boldsymbol{C(4,1) = 4}$ possible locations for the only possible pairing
    2. (2,2,0,0):
        - $C(4,2) = 6$ possible pairings
        - $C(4,2) = 6$ possible slot arrangements
        - $6 \cdot 6 = 36$ combinations possible
    - Total for $(\boldsymbol{s = 0}) \rightarrow \ \boldsymbol{4 + 36 = 40}$ ways to get $(\boldsymbol{s = 0})$
- **$s = 1$ success:**
    - (3,1,0,0):
        - $\boldsymbol{C(4,3)} = 4$ possible pairings
        - $\boldsymbol{C(4,1)} = 4$ possible locations for the pairing
        - $\boldsymbol{C(3,1)} = 3$ possible locations for the singleton
        - $\boldsymbol{4 \cdot 4 \cdot 3 = 48}$ ways to get $(\boldsymbol{s = 1})$
- **$s = 2$ successes:**
    - (2,1,1,0):
        - $\boldsymbol{C(4,2) = 6}$ possible pairs
        - $\boldsymbol{C(4,1) = 4}$ possible locations for the pair
        - $\boldsymbol{C(3,2) = 3}$ possible slot selections for the singletons
        - $\boldsymbol{2!} = 2$ possible ways to arrange the singletons in the slots
        - $6 \cdot \boldsymbol{4 \cdot 3 \cdot 2 = 144\ ways}$ to get $(\boldsymbol{s = 2})$
- **$s = 3$ successes:**
    - Impossible.
- **s = 4 successes:**
    - (1,1,1,1)
        - 4! = 24 ways to arrange the tags in the slots

- Probabilities:
  - $\boldsymbol{P(S = 0 \mid 4,4) = 40/256}$
  - $\boldsymbol{P(S = 1 \mid 4,4) = 48/256}$
  - $\boldsymbol{P(S = 2 \mid 4,4) = 144/256}$
  - $\boldsymbol{P(S = 3 \mid 4,4) = 0/256}$
  - $\boldsymbol{P(S = 4 \mid 4,4) = 24/256}$

- **Calculate L_4 when $(r_1 = 4)$:**
  Since $\boldsymbol{r_1 = 4 = N}$ we may use the recursive formula

$$L_4 = \frac{4 + P(S = 1|4,4)L_3 + P(S = 2|4,4)L_2 + P(S = 3|4,4)L_1 + P(S = 4|4,4)L_0}{1 - P(S = 0|4,4)}$$

$$L_4 = \frac{4 + \left(\frac{48}{256}\right) \times \left(\frac{51}{8}\right) + \left(\frac{144}{256}\right) \times 4}{1 - \left(\frac{40}{256}\right)} = \frac{953}{108} \approx 8.8241$$

E. For $r_1 = 5$ $(N = 4\ tags, L = 5\ slots)$

- **Probability Calculations $P(s = i \mid N = 4, L = 5)$:**
  - Total ways = $\boldsymbol{5^4 = 625}$.
  - **$s = 0$ successes:**
    1. (4,0,0,0,0):
       - $\boldsymbol{C(5,1) = 5}$ ways to position the only possible pairing
    2. (2,2,0,0,0):
       - $\boldsymbol{C(4,2) = 6}$ possible pairs
       - $\boldsymbol{C(5,2) = 10}$ possible locations for the pairs
       - $\boldsymbol{6 \cdot 10 = 60}$ ways to get the pattern (2,2,0,0,0)
    - Total for $\boldsymbol{(s = 0)\ is\ 5 + 60 = 65}$.
  - **$s = 1$ success:**
    1. (3,1,0,0,0):
       - $\boldsymbol{C(4,3) = 4}$ possible pairings
       - $\boldsymbol{C(5,1) = 5}$ possible locations for the pairing
       - $\boldsymbol{C(4,1) = 4}$ possible locations for the singleton

        - $4 \cdot \mathbf{5} \cdot \mathbf{4} = \mathbf{80}\ \boldsymbol{ways}$ **to get (s = 1)**
    - $\boldsymbol{s = 2}$ **successes:**
        1. (2,1,1,0,0):
            - $\boldsymbol{C(4,2) = 6}$ possible pairs
            - $\boldsymbol{C(5,1) = 5}$ possible locations for the pair
            - $\boldsymbol{C(4,2) = 6}$ possible slot locations for the singletons
            - $\mathbf{2!} = \mathbf{2}$ possible ways to arrange the singletons in the slots
            - $6 \cdot \mathbf{5} \cdot \mathbf{6} \cdot \mathbf{2} = \mathbf{360}\ \boldsymbol{ways}$.
    - $\boldsymbol{s = 3}$ **successes:**
        - Impossible (0 ways)
    - $\boldsymbol{s = 4}$ **successes:**
        1. (1,1,1,1,0):
            - $\boldsymbol{C(5,4) = 5}$ possible unique slot selections
            - $\mathbf{4!} = \mathbf{24}$ ways to arrange the singletons in the slots
            - $5 \cdot \mathbf{24} = \mathbf{120}\ \boldsymbol{ways}$.
              (We choose 4 slots out of 5, arrange 4 tags in them).
    - Probabilities:
        - $\boldsymbol{P(S = 0 \mid 4,5) = 65/625}$
        - $\boldsymbol{P(S = 1 \mid 4,5) = 80/625}$
        - $\boldsymbol{P(S = 2 \mid 4,5) = 360/625}$
        - $\boldsymbol{P(S = 3 \mid 4,5) = 0/625}$
        - $\boldsymbol{P(S = 4 \mid 4,5) = 120/625}$
- **Calculate $\boldsymbol{L_4^*}$ for $(\boldsymbol{r_1 = 5})$:**

$$L_4^* = 5 + \left[\left(\frac{65}{625}\right)L_4\right] + \left[\left(\frac{80}{625}\right)L_3\right] + \left[\left(\frac{360}{625}\right)L_2\right] + [0 \cdot L_1] + \left[\left(\frac{120}{625}\right)L_0\right]$$

$$L_4^* = 5 + \left(\frac{65}{625}\right)\left(\frac{953}{108}\right) + \left(\frac{80}{625}\right)\left(\frac{51}{8}\right) + \left(\frac{360}{625}\right)(4) = \frac{122009}{13500} \approx 9.0377$$

F. For $r_1 = 6$ ($N = 4$ tags, $L = 6$ slots)

- **Probability Calculations $P(s = i \mid N = 4, L = 6)$:**
  - Total ways = $6^4 = 1296$.
  - **$s = 0$ successes:**
    1. (4,0,0,0,0,0):
       - $C(6,1) = 6$ ways to position the pair
    2. (2,2,0,0,0,0):
       - $C(4,2) = 6$ possible pairs
       - $C(6,2) = 15$ possible slot arrangements for the pair
       - $6 \cdot 15 = 90$ ways.
    - Total for $(s = 0) \rightarrow 6 + 90 = 96$.
  - **$s = 1$ success:**
    1. (3,1,0,0,0,0):
       - $C(4,3) = 4$ possible pairings
       - $C(6,1) = 6$ possible locations for the pairing
       - $C(5,1) = 5$ possible locations for the singleton
       - $4 \cdot 6 \cdot 5 = 120$ ways to get $(s = 1)$
  - **$s = 2$ successes:**
    1. (2,1,1,0,0,0):
       - $C(4,2) = 6$ possible pairs
       - $C(6,1) = 6$ possible locations for the pair
       - $C(5,2) = 10$ possible unique slot selections for the singletons
       - $2! = 2$ ways to arrange the singletons in the selected slots
       - $6 \cdot 6 \cdot 10 \cdot 2 = 720$ ways to get $(s = 2)$
  - **$s = 3$ successes:**
    1. Impossible (0 *ways*)

- $s = 4$
    1. (1,1,1,1,0,0):
        - $\boldsymbol{C(6,4) = 15}$ possible unique slots selections
        - $\boldsymbol{4! = 24}$ ways to arrange the singletons in the selected slots.
        - $15 \cdot 24 = 360\ ways\ to\ get\ (s = 4)$
- Probabilities:
    - $P(S = 0|4,6) = \frac{96}{1296}$
    - $P(S = 1|4,6) = \frac{120}{1296}$
    - $P(S = 2|4,6) = \frac{720}{1296}$
    - $P(S = 3|4,6) = \frac{0}{1296} = 0$
    - $P(S = 4|4,6) = \frac{360}{1296}$

- **Calculate $\boldsymbol{L_4^*}$ for $(\boldsymbol{r_1 = 6})$:**

$$L_4^* = 6 + P(S = 0|4,6)L_4 + P(S = 1|4,6)L_3 + P(S = 2|4,6)L_2$$

$$L_4^* = 6 + \left(\frac{96}{1296}\right) \times \left(\frac{953}{108}\right) + \left(\frac{120}{1296}\right) \times \left(\frac{51}{8}\right) + \left(\frac{720}{1296}\right) \times 4 = \frac{110413}{11664} \approx 9.4661$$